PROXMOX VE CLUSTER TESTING & RESILIENCE MASTER REPORT

Complete Datacenter Physical Fault-Tolerance, Stress Failover & Storage Validation Audit

Cluster Identifier	`cisai-pve` (Proxmox VE 8.x Enterprise HA)	**Execution Date**	September 24, 2026
Cluster Nodes	`prox22` (10.31.6.172), `prox23` (10.31.6.173), `prox24` (10.31.6.174)	**Lead Engineer**	Nishu Saini / CISAI Infrastructure
Storage Subsystems	12x 20TB SAS Ceph Reef (3x Replication) + 3x 127 TiB ZFS RAIDZ2 Pools	**Overall Status**	**100% PRODUCTION SIGN-OFF (HEALTH_OK)**

22 / 22	0 Bytes	1.08s	6.46 GB/s
SCENARIOS EXECUTED	TOTAL DATA LOSS	LIVE VM FAILOVER	PEAK STORAGE READ

1. EXECUTIVE SUMMARY

On **September 24, 2026**, a comprehensive on-site physical datacenter testing program was executed on the 3-node enterprise Proxmox VE cluster. The audit covered hardware fault injection, multi-drive physical hot-pulls, 10G storage network blackouts, synthetic 1.2M IOPS live VM stress migrations, and cryptographic bitrot verification across the 127 TiB ZFS and 218 TiB Ceph storage tiers.

Final Verdict: The cluster successfully demonstrated complete fault tolerance with **zero data corruption, zero guest downtime, and 100% cryptographic SHA256 integrity**. The infrastructure is certified in **100% HEALTH_OK condition**.

2. MASTER 19-SCENARIO PHYSICAL TESTING RUNBOOK (TC-01 – TC-19)

TC ID	Test Description	Target Node	Injected Physical Fault	Result
TC-01	Single Disk Hot-Pull	`prox22`	Unlatched 20TB Ceph OSD drive in Slot 1	PASS
TC-02	Dual Disk Simultaneous Pull	`prox22`	Unlatched Slot 1 & Slot 2 drives concurrently	PASS
TC-03	3-Disk Hot-Pull	`prox22`	Unlatched Slot 1, 2, 3 drives on host	PASS
TC-04	Drive Slot Mobility Swap	`prox22`	Moved SAS drive caddy from Slot 1 to Slot 3	PASS
TC-05	Full Node Disk Removal	`prox24`	Unseated all 4 Ceph drive caddies on node	PASS
TC-06	Cross-Node Dual Disk Pull	`prox22, prox23`	Pulled 1 drive on prox22 and 1 on prox23	PASS
TC-07	Disk Hot-Pull Under Active I/O	`prox22`	Unlatched OSD drive during continuous benchmark	PASS
TC-12	ZFS Software Fault Injection	`prox24`	Simulated block device detachment on zfs-ha	PASS
TC-13	ZFS OS Boot SSD Mirror Pull	`prox23`	Unlatched Mirrored OS SSD from rear bay	PASS
TC-14	ZFS Auto-Resilver Validation	`prox23`	Re-inserted SSD; verified online resilvering	PASS
TEST-5.1	Redundant PSU Cord Pull	`prox24`	Unplugged Power Cord 1 (PSU 2 sustained load)	PASS
TC-08	Node Power Loss & HA Failover	`prox24`	Simulated catastrophic node power drop	PASS

TC ID	Test Description	Target Node	Injected Physical Fault	Result
TC-09	Dual Node Loss (Split-Brain Immunity)	`prox23, prox24`	Survivor node prox22 entered safe freeze mode	**PASS**
TC-10	10G DAC Storage Cable Pull	`prox24`	Unplugged 10G SFP+ storage interconnect cable	**PASS**
TC-11	1G Management Link Loss	`prox24`	Unplugged 1G RJ45 link; watchdog self-fenced	**PASS**
TC-15	Disk Pull During Live Migration	`prox22 -> prox24`	Pulled secondary drive during active RAM transfer	**PASS**
TC-16	Ceph min_size Protection Rule	`Cluster`	Verified pool write lock when < min_size	**PASS**
TC-17	Ceph Monitor Leader Outage	`prox24`	Stopped mon.prox24; quorum maintained (2/3)	**PASS**
TC-18	CRUSH Map Rebalance Benchmark	`Cluster`	Verified automatic weight redistribution	**PASS**
TC-19	Single Monitor SPOF Elimination	`Cluster`	Verified multi-mon Paxos election	**PASS**

3. SIMULTANEOUS 3-NODE 10G STORAGE BLACKOUT OUTAGE TEST

A high-impact total storage network blackout was executed by simultaneously disconnecting the 10G DAC interconnect cables across all three physical nodes (`prox22` , `prox23` , `prox24`) for **60 seconds**.

Metric	Target Threshold	Measured Value	Verdict
Blackout Duration	60.00 seconds	**60.00 s (Simultaneous Pull)**	PASS
Corosync Quorum During Outage	Maintain 3/3 Votes	**3/3 Votes Maintained via 1G Ring**	PASS
Ceph OSD Reconnection (TTR)	< 30.00 seconds	**14.20 s Auto-Reconnection**	PASS
Data Loss / Object Corruption	0 bytes	**0 Bytes Lost (100% Recovered)**	PASS

4. LIVE VM FIO WORKLOAD & STRESS FAILOVER BENCHMARK

A continuous direct async write stress workload was injected into **VM 9003** while executing an online live migration across nodes over the 10G interconnect.

Metric / Telemetry	Target Benchmark	Measured Result	Verdict
Peak Local 4K Random Write	High Throughput	**1,204,746 IOPS**	PASS
Live Migration Time (VM 9003)	< 30.00 seconds	**1.08 seconds (Cross-Node)**	PASS
Max Guest I/O Freeze	< 500 ms	**< 120 ms (Seamless RAM Transfer)**	PASS
Cryptographic SHA256 Integrity	0 corrupted blocks	**0 Corrupted Blocks (100% Integrity)**	PASS

5. ZFS 127 TIB STORAGE SCRUB & BITROT SELF-HEALING AUDIT

A full cryptographic block tree scrub was executed on the 7-disk RAIDZ2 pool `zfs-ha` **(127 TiB)** on `prox22` .

Verification Parameter	Pool Architecture	Error Counter Measured	Verdict
Pool State Across Nodes	7x 20TB SAS RAIDZ2	**100% ONLINE (0 Errors)**	PASS
Read Errors (READ)	7 Disks Checked	**0 Errors**	PASS
Write Errors (WRITE)	7 Disks Checked	**0 Errors**	PASS
Silent Bitrot (CKSUM)	Fletcher4 / SHA256	**0 Checksum Errors**	PASS

6. DD IOPS & THROUGHPUT BENCHMARK AUDIT (ZFS VS CEPH)

Payload Size	ZFS RAIDZ2 Storage Pool ('zfs-ha')		Ceph Distributed Storage (RBD 3x Replica)	
	Write (bs=1M)	**Read (bs=1M)**	**Write (bs=1M)**	**Read (bs=1M)**
10 MiB	199.4 MB/s	167.2 MB/s	68.4 MB/s	113.6 MB/s
20 MiB	385.2 MB/s	331.1 MB/s	93.8 MB/s	170.2 MB/s
50 MiB	877.8 MB/s	797.5 MB/s	105.8 MB/s	255.2 MB/s
100 MiB	1.49 GB/s	1.43 GB/s	113.7 MB/s	298.9 MB/s

Payload Size	ZFS RAIDZ2 Storage Pool ('zfs-ha')		Ceph Distributed Storage (RBD 3x Replica)	
	Write (bs=1M)	Read (bs=1M)	Write (bs=1M)	Read (bs=1M)
200 MiB	2.29 GB/s	2.57 GB/s	111.7 MB/s	336.1 MB/s
500 MiB	3.50 GB/s	4.78 GB/s	104.4 MB/s	334.3 MB/s
1000 MiB (1 GB)	**4.59 GB/s**	**6.46 GB/s**	**98.5 MB/s**	**243.9 MB/s**

7. TECHNICAL ROOT CAUSE ANALYSIS (RCA) SUMMARY

- **Drive Hot-Pull SCSI Desync:** Hot-swapping physical SAS drives caused Linux SCSI mid-layer re-enumeration (`/dev/sdb` [8:16] ➡️ `/dev/sdn` [8:208]), leaving LVM device-mapper pointing to stale minor handles. Formulated standard automated SOP (`vgchange -an` , `udevadm trigger` , `vgchange -ay` , `ceph-volume lvm activate --all`) to instantly restore OSDs without rebooting.
- **Web UI Transition Artifacts:** Resolved HA maintenance mode wrench and `?` icons by restoring `pvestatd` metric broadcast daemons across nodes.

8. FINAL PRODUCTION SIGN-OFF

Cluster Quorum	🟢 3 / 3 Votes Quorate (YES)	**Ceph Health**	🟢 12 / 12 OSDs UP & IN (HEALTH_OK)
ZFS Pools	🟢 3 / 3 Pools ONLINE (0 Errors)	**HA VMs**	🟢 VMs 9001, 9002, 9003, 9004 Started
Final Sign-Off	**CERTIFIED PRODUCTION READY — ALL SYSTEMS OPERATIONAL**		